

## Quiz 5

M 375T Mathematical Foundations of Machine Learning — Fall 2026

Optional · 15 minutes · closed book · no devices

Name: \_\_\_\_\_ UT EID: \_\_\_\_\_

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1.

[5 pts]

```
import numpy as np
W = np.array([[0, 2, 0, 1],
              [2, 0, 1, 0],
              [0, 1, 0, 1],
              [1, 0, 1, 0]], dtype=float)
d = W.sum(axis=1)
L = np.diag(d) - W
v = np.array([1., 1., -1., -1.])
print(v @ L @ v)                                # line 1
```

- (a) Write  $d$  and  $L$ .
- (b) Predict line 1 using the Dirichlet form  $v^T L v = \frac{1}{2} \sum_{i,j} W_{ij} (v_i - v_j)^2$ . Which edges contribute, and how much each?
- (c) For  $A = \{0, 1\}$  and  $B = \{2, 3\}$  compute  $\text{cut}(A, B)$ ,  $\text{vol}(A)$ ,  $\text{vol}(B)$  and  $\text{Ncut}(A, B) = \text{cut}(A, B) \left( \frac{1}{\text{vol}(A)} + \frac{1}{\text{vol}(B)} \right)$ . How is  $\text{cut}(A, B)$  related to line 1?

Answer:

2. **A walk that never settles, and the fix.** Vertices  $0, 1, 2$ ; the only edges are  $0-1$  with weight 2 and  $1-2$  with weight 1. The walker moves with the rule  $P_{ij} = W_{ij}/d_i$ . [5 pts]
- (a) Write  $d$ ,  $P$ , and the invariant measure  $\pi$  (a probability vector).
  - (b) The walker starts at vertex 1, so  $\mu_0 = (0, 1, 0)^T$ . Compute  $\mu_1$  and  $\mu_2$ . What does the law do as  $t$  grows, and why does this happen although the graph is connected? Name the eigenvalue of  $P^T$  and compute its eigenvector.
  - (c) Write the lazy transition matrix  $Q = \frac{1}{2}(I + P)$  and compute the lazy law after one step from vertex 1. Compare it with  $\pi$ . What is the lazy law after two steps?

Answer:

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Optional quiz, can only help. Total: 10 points. Lowest two quiz scores are dropped at the end of the semester.